

1404. Number of Steps to Reduce a Number in Binary Representation to One

Solved 

Medium

Topics

Companies

Hint

Given the binary representation of an integer as a string `s`, return *the number of steps to reduce it to 1* under the following rules:

- If the current number is even, you have to divide it by 2.
- If the current number is odd, you have to add 1 to it.

It is guaranteed that you can always reach one for all test cases.

Example 1:

Input: `s = "1101"`

Output: 6

Explanation: "1101" corresponds to number 13 in their decimal representation.

Step 1) 13 is odd, add 1 and obtain 14.

Step 2) 14 is even, divide by 2 and obtain 7.

Step 3) 7 is odd, add 1 and obtain 8.

Step 4) 8 is even, divide by 2 and obtain 4.

Step 5) 4 is even, divide by 2 and obtain 2.

Step 6) 2 is even, divide by 2 and obtain 1.

Example 2:

Input: `s = "10"`

Output: 1

Explanation: "10" corresponds to number 2 in their decimal representation.

Step 1) 2 is even, divide by 2 and obtain 1.

Example 3:

Input: `s = "1"`

Output: 0

Constraints:

- `1 <= s.length <= 500`
- `s` consists of characters '0' or '1'
- `s[0] == '1'`

Python:

```
class Solution:
    def numSteps(self, s: str) -> int:
        steps = 0
        carry = 0

        for i in range(len(s) - 1, 0, -1):
            if (((ord(s[i]) & 1) + carry) & 1):
                steps += 2
                carry = 1
            else:
                steps += 1

        return steps + carry
```

JavaScript:

```
const numSteps = s => {
    let steps = 0, carry = 0;

    for (let i = s.length - 1; i > 0; i--) {
        if (((s.charCodeAt(i) & 1) + carry) & 1) {
            steps += 2;
            carry = 1;
        } else
            steps += 1;
    }

    return steps + carry;
};
```

Java:

```
class Solution {
    public int numSteps(String s) {
        int steps = 0, carry = 0;

        for (int i = s.length() - 1; i > 0; i--) {
            if ((s.charAt(i) & 1) + carry == 1) {
                steps += 2;
                carry = 1;
            } else
                steps += 1;
        }

        return steps + carry;}}}
```